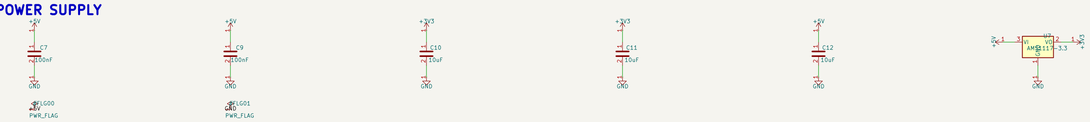
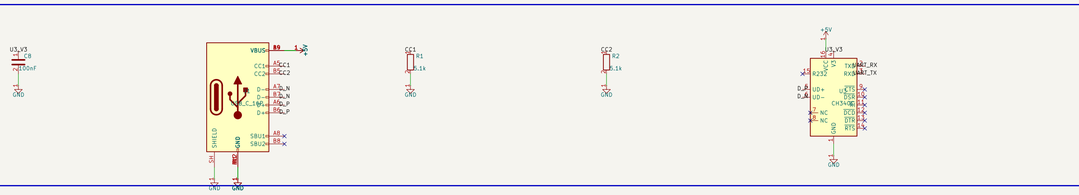
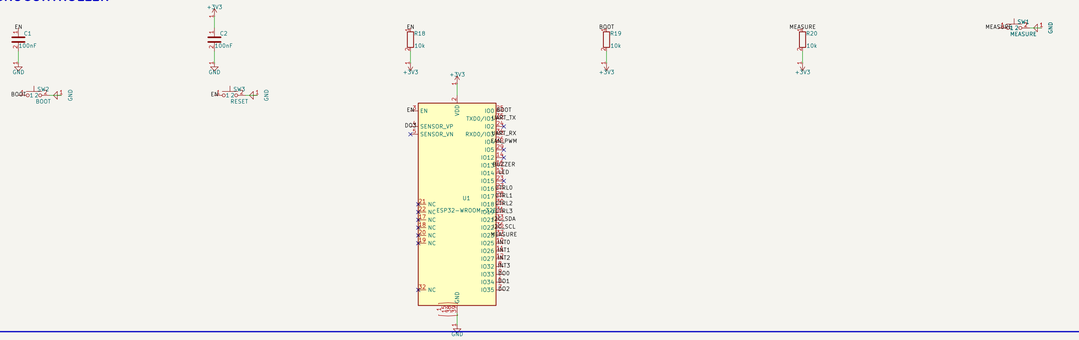
POWER SUPPLY

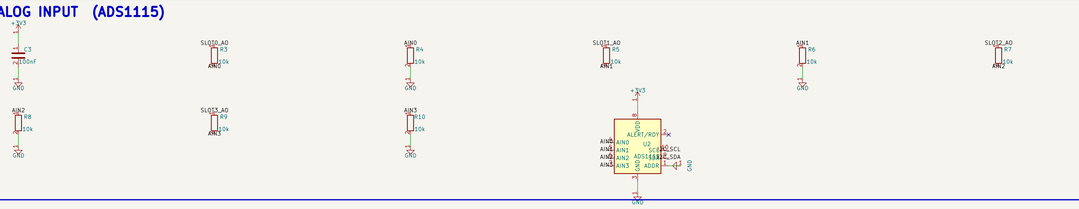




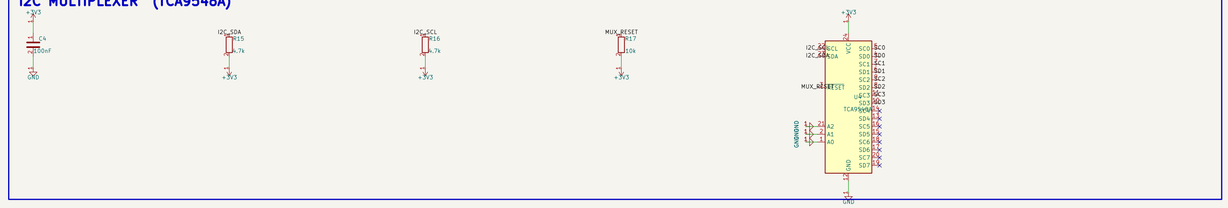
MICROCONTROLLER



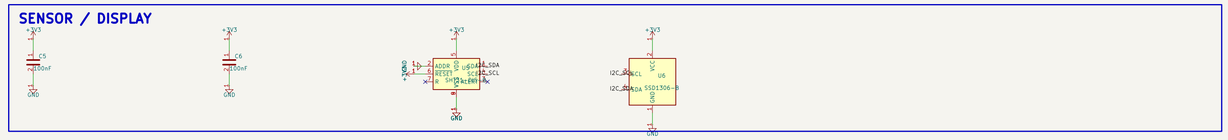
ANALOG INPUT (ADS1115)



I2C MULTIPLEXER (TCA9548A)



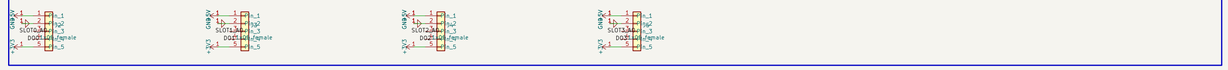
SENSOR / DISPLAY



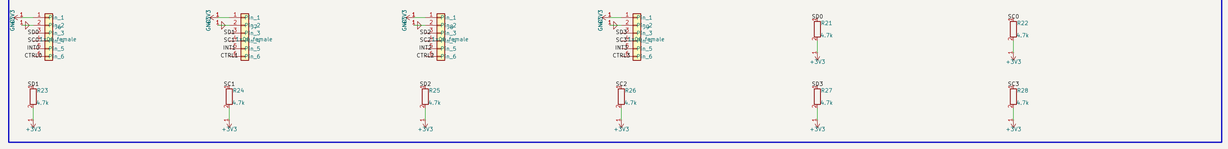
OUTPUTS



ANALOG SLOTS



DIGITAL SLOTS



NOTES

1. +3V3 is generated by U7 (AMS1117CD-3.3) from the USB-C +5V rail; every IC has a 100nF decoupling cap at its supply pin.
2. ANALOG SLOTS J2..J5 = 1:+5V 2:GND 3:AO 4:DO 5:3V3 (MQ-type module seats on pins 1-4).
3. DIGITAL SLOTS J6..J9 = 1:3V3 2:GND 3:SDA 4:SCL 5:INT 6:CTRL (each slot is one TCA9548A channel).
4. I2C ADDRESSES ADS1115 0x48 TCA9548A 0x70 SHT31 0x44 OLED 0x3C.
5. S1 = MEASURE (start a sniff cycle) S2 = BOOT S3 = RESET. Manual flashing, no auto-reset transistors.
6. Connectivity is carried by net-name labels; the authoritative pin/net model is electrical/board.py (validated, ERC-clean).
7. NC = not connected (no-connect flag placed). All dimensions mm; passives 0603 unless noted.

Connectivity is net-name based: authoritative model = board.py
Controller, 4 analog + 4 digital sensor slots

OpenSmell		
Sheet: /		
File: smell_monitor_v1_1.kicad_sch		
Title:		
Size: A1	Date: 2026-10-16	**Rev: A**
KiCad E.D.A. 10.0.4		Id: 1/1